

Department of Mining Engineering
IIT(ISM), Dhanbad
Class Test
Semester: IV Sem (ME) Session 2022-23 (WS)
Subject: Underground Coal Mining (MNC 208)

Time 40 minutes

Marks : 20

Qn. No.	Question	Marks
1	Explain how two legged shield supports offer better stability at longwall face?	4
2 <i>x</i>	How do you calculate average support density / mean load density at maximum and minimum span of a longwall face? Assume your own data, if required.	5
3 <i>x</i>	Why debris on top of the canopies of longwall powered supports should be cleaned at regular intervals?	3
4 <i>x</i>	Using bulking factor theory, calculate the <u>caving height</u> of <u>caving zone</u> of a longwall face with 4 meter height of extraction.	4
5	Give two reasons why wider longwall faces are preferred.	4

Mineralogical, etc.

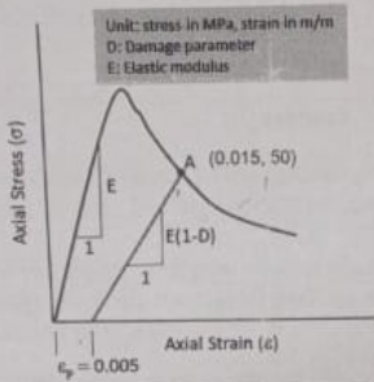
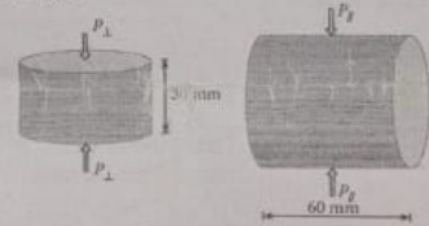
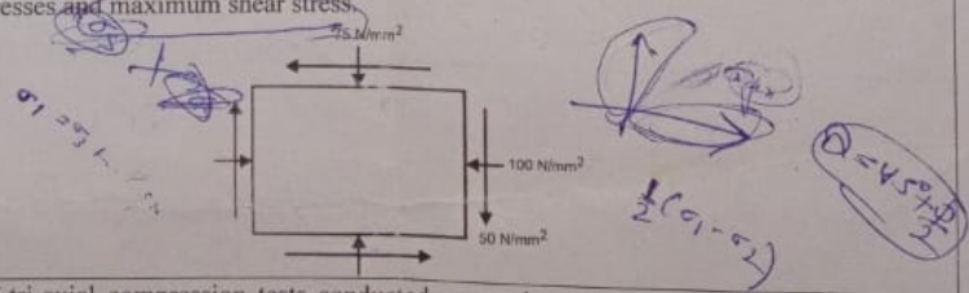
Mid Semester Examination (Winter 2022-2023)

Rock Mechanics (MNC 205)

Total Time: 8:30 AM to 10:30 AM (2 Hours)

Maximum Marks: 32

IV Semester B.Tech. (Mining Engineering)

Q. No	Question	Mark
1	The uniaxial stress-strain behavior of a rock sample is shown in the figure. The elastic modulus E is 20,000 MPa. If ϵ_p is the plastic strain, find the value of the damage parameter D (rounded off to two decimal place).  Unit: stress in MPa, strain in m/m D: Damage parameter E: Elastic modulus	3
2	Point load tests were carried out on two sedimentary rock specimens of 54mm diameter (NX core), as shown the following figure. The loads $P_{\perp}$ and $P_{\parallel}$ at failure are 6.28 kN and 4.71 kN, respectively. Find the point load strength index $I_s(50)$ in the two directions and compute the point load strength anisotropy index $I_a(50)$. 	3
3	Explain the principle of the overcoring method of in-situ stress determination. Compare the common methods in determination of in-situ stress by overcoring technique.	3+3
4	Differentiate between plane stress and plane strain conditions and indicate their significance in rock engineering.	3+1
5	The state of stress in two-dimensionally stressed body at a point is shown in figure. Determine the principal stresses and maximum shear stress. 	5
6	A series of tri-axial compression tests conducted on sandstone samples reveal the following relationship between major and minor principal stresses: $\sigma_1 = 50 + 3\sigma_3$. Determine the angle of internal friction in degrees and cohesion in MPa of sandstone.	4
7	a) Discuss briefly four applications of rockmass classification systems in mining and rock engineering projects. b) The measurements of a site investigation are as follows: $J_n = 8$, $J_r = 1.3$, $J_a = 2.0$, $J_w = 1.0$, $SRF = 2.3$. The RQD values taken from five boreholes of the area are 95.2, 92.3, 96.3, 94.3 and 97.0 respectively. Determine the rock mass quality index (Q).	4+3=7

Fuel Minerals and Metallurgical Engineering Department
IIT (ISM) DHANBAD

Mid Sem; IVth B.Tech
Subject: Introduction Fuel Technology (FME 222)

Winter Session: 2022-23
Time: 2 Hrs Max Marks: 60

Q.No.	Questions	Marks
1.a)	Write the significance and importance of the following- <u>LTGK, Stamping, Rank of coal, Caking Index, Metamorphism.</u>	(10)
b)	What is the role of <u>proximate analysis</u> in coal characterization? What information can be inferred from this analysis? Write the significance of various constituents of coal and how they can affect any process and production.	(10)
2.a)	Justify the following- (i) <u>Mineral matter is always 10% higher than the ash content of coal.</u> (ii) <u>Net calorific value is always lower than the gross calorific value.</u>	(06)
b)	What is <u>Metaplast theory</u> ? How the formation of coal to coke takes place at an elevated temperature? Explain.	(08)
c)	<u>Draw and discuss in brief a coal handling unit of a coke oven plant with the help of a flowsheet.</u>	(06)
3. a)	What is <u>water equivalent</u> in bomb calorimeter? <u>Explain different correction factors</u> and their importance during calorific value determination.	(10)
b)	What is <u>coke reactivity</u> ? Calculate the CRI and CSR value of <u>non-reactive coke</u> with zero percent abrasion.	(04)
c)	Given coal sample consists of Moisture= 3%, Ash= 25%, Volatile Matter= 20% Calculate – i) the analysis on daf & pure coal basis ii) <u>GCV</u> and UHV of the coal iii) Mineral matter present in the coal	(06)

DEPARTMENT OF MINING ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES), DHANBAD

Mid-Semester Examination

Examination: IV B. Tech. (Mining Engineering)

Session: 2022-23

Semester: Winter

Subject: Mine Ventilation (MNC 206)

Time: 2 Hours

Instruction: Answer ALL questions

Total Marks: 60

Q. No.	Question	Marks
1.	Discuss the different sources of heat in underground coal mines. OR With the help of a schematic diagram and superimposed pressure-enthalpy-state diagram, explain the constructional features and working of a vapour-compression refrigeration system used in mine air-conditioning.	15
2.	Explain the different modes of methane emission into underground coal mine atmosphere, and side by side discuss how to control such emission.	15
3.	Discuss about the generation of radon gas, its physiological effects and control measures in the mine environment.	10
4.	The analysis of air sample collected from an underground coal mine revealed the following: $O_2 = 16.52\%$, $N_2 = 77.93\%$, $CO_2 = 3.1\%$, $CH_4 = 2.44\%$, $CO = 0.01\%$. Calculate the percentage of (i) Atmospheric air, (ii) Blackdamp, (iii) Firedamp, and (iv) Whitedamp in the mine air.	10
5.	The airflow rate in an <u>ascensionally ventilated</u> underground coal mine airway of 4.0 m width and 3.0 m height is $30 \text{ m}^3/\text{s}$. If the rate of methane emission into the airway is $7.5 \text{ m}^3/\text{s}$, calculate the methane layering number and critically analyse whether it is adequate to break the methane layering in the airway or not. What measures would you take to break the methane layering in this situation?	10

$$\left(\frac{m}{s^2} \times \frac{m^3}{s} \right)^{1/3} = \text{App } m/s$$

$$\frac{9.81 \times 1}{5} = \frac{(m^3/s)}{s} \Rightarrow 1.962 = \frac{m^3/s}{s}$$

Department of Mining Engineering
IIT(ISM), Dhanbad

IV Semester B. Tech (Mining Engineering)

Session 2022-23

Mid-Semester Examination

Subject: Underground Coal Mining (MNC 208)

Time – 2 hours

Full marks - 60

All questions are compulsory
Questions

Question No.	Questions	Marks
1.	a) What do you understand by in-seam mining? In which conditions in-seam mining is preferred? Describe in brief, the advantages and disadvantages of horizon mining.	2+3+5 (10)
2.	Compare different underground coal mining methods like Bord & Pillar mining, Room & Pillar mining and Longwall mining with respect to the following factors: a) % age of extraction b) Effectiveness of strata control c) Risk of spontaneous heating d) Mass production capability e) Fleeting of face machinery	5x5 (25)
3.	Write short notes on the following: a) Breaker line support b) Hazards of shaft sinking c) Diagonal line of extraction (5)	5 x 3 (15)
4.	Determine the size of Bord and Pillar depillaring panel of a coal mine which can be extracted safely within the incubation period of 12 months with a rate of production of 500 Tonne per day. Thickness of the coal seam is 4.5 meter. Assume your own conditions as necessary.	10

INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES), DHANBAD

Exam: IV B. Tech (Mining Engineering)

Session: 2022-23

Semester: Winter

Subject: MINE VENTILATION (MNC 206)

Total Marks: 100

Time: 3 hours

Instruction: Answer any FIVE questions

Q. 1	a) Discuss the generation sources of <u>NO_x</u> in underground mine atmosphere and its physiological effects on the human being. b) A 2.5 m thick gassy coal seam is occurring at a depth of 500 m from the surface. With neat sketches, explain the direct method of measurement of gas content of the coal seam.	3 + 3 14
Q. 2	a) Discuss the method of sampling of air in underground mine atmosphere and its analysis using a Gas Chromatograph. b) A 2000 m long and 200 m wide longwall panel is being worked by retreating method. With a neat sketch, discuss the methane drainage technique using surface boreholes for extraction of methane accumulated in the goaf of the longwall panel.	4 + 6 10
Q. 3	a) Discuss the applications of booster fan and auxiliary fan. b) A mine has three splits A, B and C, each of resistance $0.1 \text{ N s}^2/\text{m}^8$. The trunk airways have a total resistance of $0.2 \text{ N s}^2/\text{m}^8$. If the desired air quantity in splits A and B is $25 \text{ m}^3/\text{s}$ and $15 \text{ m}^3/\text{s}$, respectively, calculate the airflow in split C when the surface fan generates a pressure of 700 Pa. Also, determine the specifications of the flow control devices in splits A and B to achieve these specified air quantities.	3 + 3 14
Q. 4	a) With the help of a sketch, critically analyze how the change in direction of airflow due to natural ventilation influences the operation of main mine ventilation fan. b) A mine fan generates 500 Pa pressure and circulates $25 \text{ m}^3/\text{s}$ of air through the mine consisting of two splits, A and B. Split A and B circulates $15 \text{ m}^3/\text{s}$ and $10 \text{ m}^3/\text{s}$ of air, respectively. It is desired to increase the quantity of airflow in split B from $10 \text{ m}^3/\text{s}$ to $15 \text{ m}^3/\text{s}$ by installing a booster fan in it. Calculate the size of the booster fan if the resistance of the shafts and trunk airways is $0.2 \text{ N s}^2/\text{m}^8$.	3 + 5 12
Q. 5	a) With the help of a schematic mine ventilation network, discuss the purpose of the following ventilation control devices: (a) Regulator, and (b) Air-crossing. b) A horizontal underground mine gallery of 3.5 m wide, 2.5 m high and 700 m long lying at a depth of 250 m is naturally ventilated by means of two vertical shafts of 5.0 m diameter each. The gallery contains a normal bend with deflection of $\pi/3$ radians and radius of curvature of 3 m. The temperature measurements in the shafts are as follows: Downcast shaft-top = 293 K, Downcast shaft-bottom = 296 K Upcast shaft-top = 303 K, Upcast shaft-bottom = 304 K The coefficient of friction k for the shafts is $0.01 \text{ N s}^2/\text{m}^4$ and for the underground gallery is $0.02 \text{ N s}^2/\text{m}^4$. If the average barometric pressure in the shafts is 101.33 kPa, calculate the velocity of airflow in the gallery due to natural ventilation. Neglect the shock loss in the shafts.	6 14
Q. 6	a) With a neat sketch discuss the constructional features and working of an axial-flow fan, and mention its merits and limitations over the centrifugal fan. b) With a neat sketch, briefly discuss the mine characteristic curve, P-Q characteristic curve of fan, and operating point of fan.	8 + 4 2 + 6

Department of Mining Engineering
IIT(ISM), Dhanbad

IV Semester B. Tech (Mining Engineering)

Semester Examination

Session 2022-23

Subject: Underground Coal Mining (MNC 208)

Time – 3 hours

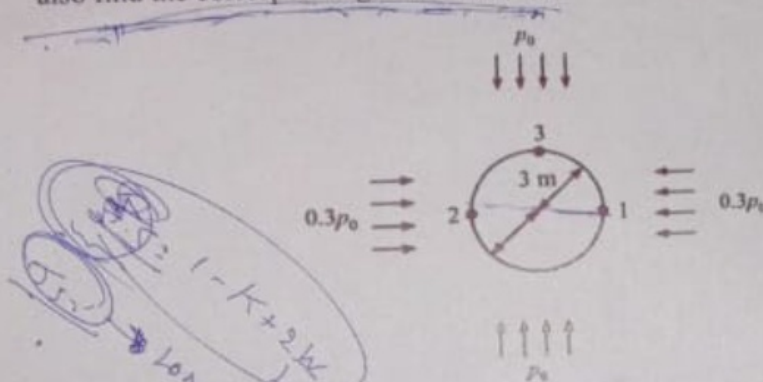
Full marks - 100

All questions are compulsory

Question No.	Questions	Marks
1.	a) With the help of neat labeled sketch show the layout of a longwall mine with three entry panel system, showing the following: I. One longwall panel of 3 km long and 300 m wide at retreat of 300 m from the installation chamber II. One extracted longwall panel adjacent to panel being extracted III. One developed longwall panel of 3 km long and 300 m wide, and IV. Three headings of a longwall panel being developed adjacent to the developed panel b) Show the main gate and tail gate roads with their ventilation status and quantity of air in each gate road of the longwall panel under extraction producing 8000 tonne per day c) Show the position of gate belt conveyor, Energy Train, Face equipment in the working longwall panel d) Show the ventilation control devices like ventilation stoppings, regulators, isolation stoppings etc. in the longwall panel e) Give the list of equipment in longwall panel and development headings f) Calculate the cycle time of the longwall operating face of 300 m wide, assuming your own data as required.	10+4+4 +4+4+4
2.	a) With the help of neat labeled sketch explain the steps of extracting a coal pillar of 40 m X 40 m centre to centre distance by Split & Fender method. b) Show the support system in original galleries, split galleries and goaf edges.	10+5

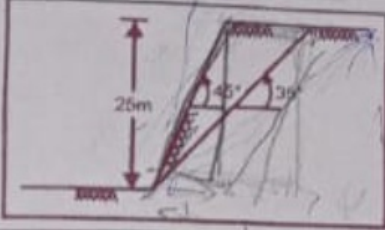
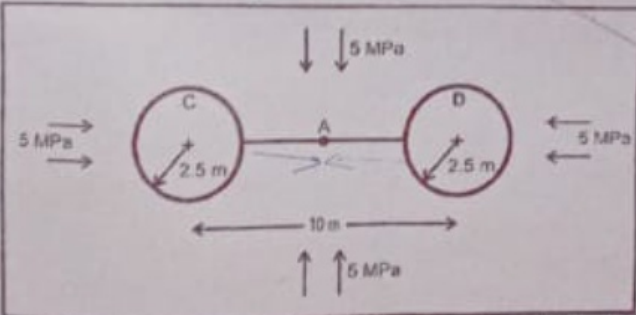
Time: 3 Hours

Instructions: 1) Answer any FIVE questions 2) Assume suitable data wherever necessary

Q. No	Question	Marks									
1.	a) At a depth of 450 m, a 3 m diameter circular tunnel is driven in rock having unit weight of 26 kN/m³ and uniaxial strength and tensile strengths of 40.0 MPa and 2.0 MPa, respectively. If the stress field ratio is 0.3. Find the induced stress at points (1), (2) and (3) as shown in the figure and also find the corresponding factor of safety.  (b) The stress in a granitic rock mass has been measured by the hydraulic fracturing technique. Two tests are conducted in a vertical borehole: one test at a depth of 800 m and the other test at a depth of 1200 m. The results are as follows: <table border="1"> <thead> <tr> <th>Depth (m)</th><th>Breakdown pressure, P_B (MPa)</th><th>Shut-in pressure, P_s (MPa)</th></tr> </thead> <tbody> <tr> <td>800</td><td>14.0</td><td>7.0</td></tr> <tr> <td>1200</td><td>24.5</td><td>19.0</td></tr> </tbody> </table> Given that the tensile strength of the rock is 10 MPa, estimate and list the values of σ_1, σ_2, and σ_3 at the two depths. Average unit weight of the rock mass is 26 kN/m³.	Depth (m)	Breakdown pressure, P_B (MPa)	Shut-in pressure, P_s (MPa)	800	14.0	7.0	1200	24.5	19.0	10
Depth (m)	Breakdown pressure, P_B (MPa)	Shut-in pressure, P_s (MPa)									
800	14.0	7.0									
1200	24.5	19.0									
2.	(a) Explain briefly the applications of Geomechanics classification of rock masses and Rock Mass Quality Index of rock mass classification systems in mining and rock engineering. (b) In a 1500-mm rock core run, the following rock pieces were recovered from a borehole: 53 mm, 108 mm, 125 mm, 75 mm, 148 mm, 320 mm, 68 mm, 145 mm, 35 mm and 134 mm. Find the RQD and the core recovery ratio. (c) A fine-grained soil is found to have a liquid limit of 90% and a plasticity index of 50. The natural water content is 28%. Determine the liquidity index and indicate the probable consistency of the natural soil.	5+									
3.	(a) The RMR of immediate roof of a coal mine is estimated to be 42. The stress conditions at a given time in the roof are as follows: $\sigma_1 = 25 \text{ MPa}, \sigma_3 = 5 \text{ MPa}$ Considering m_i is 10 and uniaxial compressive strength (σ_{ci}) is 50 MPa. The coal mine galley is created by machine cutting i.e., completely avoiding blasting operations. Does this roof stratum fail? Explain your answer through calculations. (b) Discuss briefly the methods of reducing subsidence influence in mines.	10									

Unit

$$W = \frac{\gamma h}{2} [H]$$

Q. No	Question	Marks
4.	(a) Explain the general procedure for monitoring of an underground excavation stability in hard rock using geotechnical instrumentation. (b) An opencast mine bench has a potential failure plane as indicated below. The unit weight, cohesion and angle of internal friction of the rock mass are 24.5 kN/m³, 0.02 MPa and 30° respectively. Calculate the driving force, on the potential failure plane and determine the factor of safety of slope under given condition. 	10 10
5.	(a) Draw the principal modes of failure of mine pillars and also illustrate the evolution of fracture and failure in a pillar in massive rock. (b) A flat-lying coal seam 3.1 m thick and 80 m below ground surface has been mined with 5.0 m rooms and 7.0 m square pillars, over the lower 2.2 m of the seam. Determine the factor of safety of the pillars, and assess the feasibility of stripping an extra 0.6 m of coal from the roof. The strength of the square pillars, of width W_p and height h, is given by $S = 7.5h^{-0.66} w_p^{0.46}$ where S is in MPa, and h and W_p are in m. The unit weight of the overburden rock is 26 kN/m³. (c) In a hydrostatic stress field, point A is in the middle of the two circular openings as shown in the figure. Determine the radial stress at point A. 	5 10 5
6.	(a) Draw the radial support pressure-displacement curves for the rock mass and the support system (rock-support interaction diagram) of a tunnel. Explain briefly the influence of support stiffness and of the timing of its installation on support performance in a tunnel. (b) At a point in a rock structure, the stresses on two mutually perpendicular planes are 35 MN/m² (tensile) and 15 MN/m² (tensile). The shear stress across these planes is 9 MN/m². Find the magnitude and direction of the resultant stress on a plane making an angle of 40° with the plane of first stress. Find also the normal and tangential stresses on the planes. (c) Differentiate between Finite Element Method and Finite Difference Method.	4+4 3+3 6

Note-1: Use the following formulas (Kirsch equations) wherever necessary

$$\sigma_{rr} = \frac{p}{2} \left[(1+K) \left(1 - \frac{a^2}{r^2} \right) - (1-K) \left(1 - 4\frac{a^2}{r^2} + \frac{3a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{\theta\theta} = \frac{p}{2} \left[(1+K) \left(1 + \frac{a^2}{r^2} \right) + (1-K) \left(1 + \frac{3a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{r\theta} = \frac{p}{2} \left[(1-K) \left(1 + \frac{2a^2}{r^2} - \frac{3a^4}{r^4} \right) \sin 2\theta \right]$$